GAME LIBRARY MANAGER

Linux Shell Scripting Project Report

Project: L76 – Game Library Manager

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Environment: Ubuntu Linux / Bash

Project Folder: `~/linux_project/gamelib`

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1. Introduction and Objectives

Game Library Manager is a menu-driven Linux shell scripting application designed to manage a user's video game collection.

It stores game details such as title, platform, genre, hours played and completion status.

Objectives

- Develop a menu-driven Bash application.
- Add and manage game records.
- Filter games by platform or genre.
- Calculate total hours played.
- Calculate game completion percentage.
- Display the top five most played games.
- Create compressed backups.
- Maintain an activity log.
- Generate weekly reports using Cron scheduling.

2. Project Folder Structure

```
gamelib/
├── main.sh
├── README.md
├── lib/
│   └── functions.sh
├── data/
│   └── games.csv
```

```
    ├── reports/
    ├── backups/
    └── logs/
        └── activity.log
```

`main.sh` provides the main menu. `functions.sh` contains the feature functions. The data directory stores game records, reports stores generated summaries, backups stores compressed backups, and logs records program activities.

3. Feature 1: Add a Game

The user can enter the following information:

- Game Title
- Platform
- Genre
- Hours Played
- Status

The program validates the input and saves the new game record in `games.csv`.

Screenshot: Feature 1 – Add a Game

[Insert actual Ubuntu terminal screenshot here]

4. Feature 2: Filter Games by Platform or Genre

Users can search their game collection using a platform or genre.

The program searches the CSV file without considering uppercase or lowercase differences and displays matching records.

Screenshot: Feature 2 – Filter Games

[Insert actual Ubuntu terminal screenshot here]

5. Feature 3: Show Total Hours Played

This feature uses the `awk` command to calculate the total hours played across all games.

The program skips the CSV header and adds the hours from each game record.

Screenshot: Feature 3 – Total Hours Played

[Insert actual Ubuntu terminal screenshot here]

6. Feature 4: Show Completion Percentage

This feature calculates the percentage of completed games.

Formula:

```
Completion Percentage = (Completed Games / Total Games) × 100
```

The program displays the number of completed games and the completion percentage.

Screenshot: Feature 4 – Completion Percentage

[Insert actual Ubuntu terminal screenshot here]

7. Feature 5: Top 5 Most Played Games

This feature sorts game records according to hours played and displays the five games with the highest playing time.

Linux commands such as `sort`, `tail` and `head` are used.

Screenshot: Feature 5 – Top 5 Most Played Games

[Insert actual Ubuntu terminal screenshot here]

8. View Data and Backup

The View Data option displays the game records stored in `data/games.csv`.

The Backup option creates a compressed `.tar.gz` archive of the data directory using the Linux `tar` command.

Screenshot: View Data and Backup

[Insert actual Ubuntu terminal screenshot here]

9. Activity Log and Input Validation

The activity log records important application activities, including:

- Adding a new game.
- Viewing game statistics.
- Filtering game records.
- Creating backups.

- Generating weekly reports.

Input validation checks that required fields are not empty, hours are numeric, and game status is valid.

Screenshot: Activity Log

[Insert actual Ubuntu terminal screenshot here]

10. Cron Task and Weekly Report

The project is designed to generate a weekly game library summary every Sunday at 8:00 PM.

Cron expression:

```
0 20 * * 0
```

The scheduled task executes:

```
./main.sh --cron
```

The generated report is saved in the `reports/` directory.

Screenshot: Cron Configuration and Generated Report

[Insert actual screenshots after testing Cron here]

11. What I Learned

Through this project, I practised Bash scripting, functions, loops, conditional statements, case statements and input validation.

I learned how to manage CSV files, calculate statistics using `awk`, filter and sort records, create compressed backups using `tar`, maintain activity logs and schedule automated tasks using Cron.

Conclusion

Game Library Manager demonstrates how Linux shell scripting can be used to develop a practical menu-driven application.

The project provides features for managing game records, filtering information, calculating statistics, creating backups and generating scheduled reports.

It helped me improve my understanding of Ubuntu Linux, Bash scripting and Linux command-line utilities.